Installation Manual

Vertical lifting fabric door
ASSA ABLOY VL3110 Megadoor

ASSA ABLOY

ASSA ABLOY Entrance Systems

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door opening solutions



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1. Introduction

1.1 General information

The installation instructions for the ASSA ABLOY VL3110 Megadoor vertical lifting fabric door are compiled in conformance with European standard EN 12635:2002 + A1:2008 “Industrial, commercial and garage doors – Installation and use”. The door and its components shall be fitted in compliance with these installation instructions.

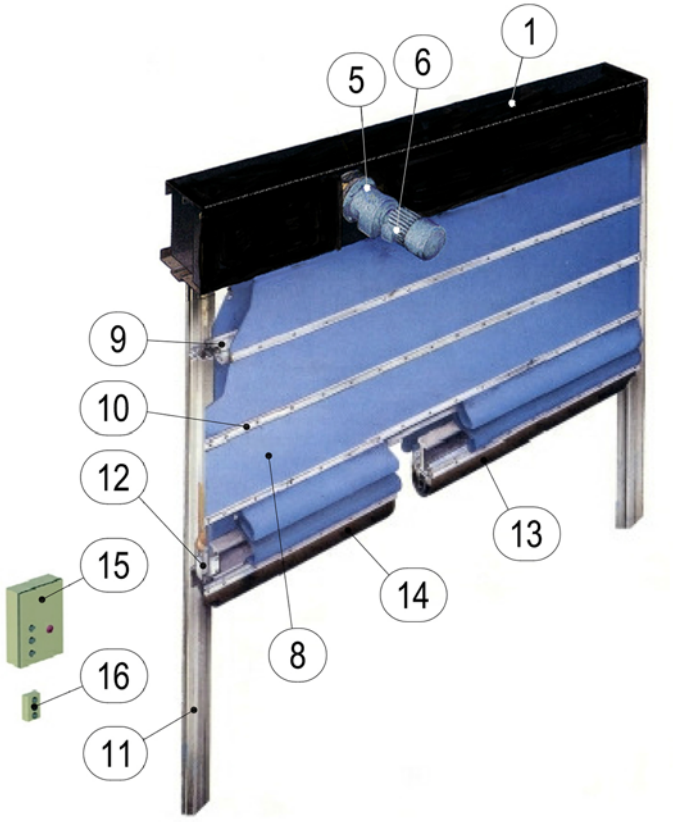
Pay special attention what is written about mounting under the headings “Installation - general”, “Guide rails - mounting surfaces” and “Header box - mounting surfaces”.

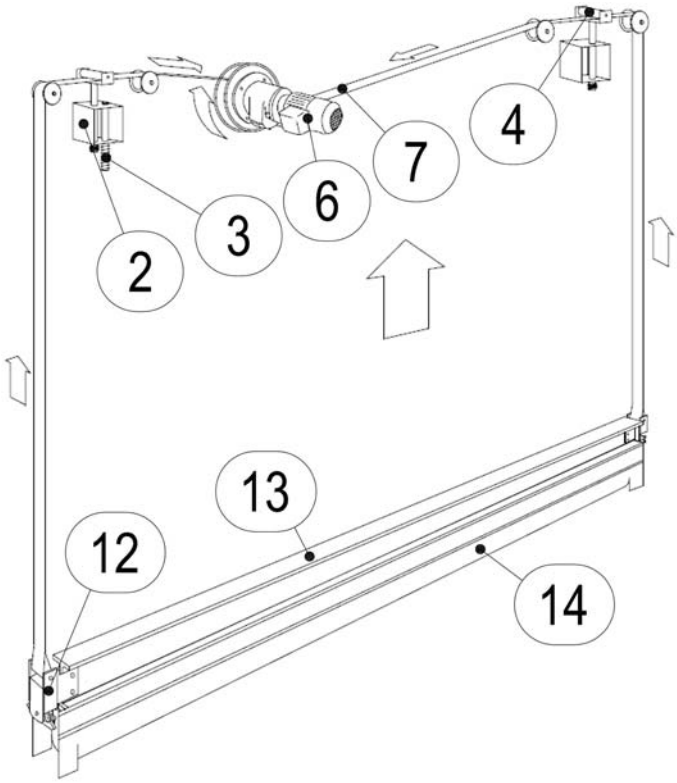
Instructions according to EN 12604:2000 “Mechanical aspects – requirements” and EN 12453:2000 “Safety in use of power operated doors - requirements” must be followed and verified if necessary.

The door is intended for installation by a professional installer. Installation carried out by anyone other than professional installers could result in a non conforming door and also risk of personal injury.

Personnel who have limited training may assist the installer.

1.2 Door leaf and header box parts list





1) Header box
2) Limit switch box
3) Push rod
4) Pull rod
5) Gear motor
6) Brake lever
7) Lifting belt
8) Door leaf
9) Intermediate section
10) Clamp strip
11) Guide rail
12) Safety arrester*
13) Bottom section
14) Floor seal
15) Control unit
16) Push button unit

* The safety arrester, made of stainless steel, is activated during each opening sequence and also locks the door in the closed position.

1.3 Delivery

An ASSA ABLOY VL3110 Megadoor vertical lifting fabric door is usually delivered in two packages:

1)
- Header box including door control, electrically connected and test run
- Drive unit (lying)
- Door leaf
- Bottom section
- Fittings, screws, etc, in box
- Motor casing, if any

2)
- Guide rail / jambs



The image shows a delivery including motor cover and guide rails with jambs.

2. Installation

2.1 General

1) The door is installed in the building by screw fitting.
2) Screws are to be torque tightened according to the table below (± 5%):

Taptite (self-tapping) 8.8:

Thread	Drill diameter	Material thickness	Torque
M5	4.5 mm	1.5 -2.5 mm	6.5 Nm
M6	5.5 mm	2.5 - 4.0 mm	11.0 Nm
M8	7.4 mm	4.0 - 6.5 mm	26.5 Nm
M10	9.3 mm	6.5 - 10.0 mm	52.5 Nm

Normal screw 8.8:

Thread	Drill diameter	Torque
M8	6.8 mm	24 Nm
M10	8.5 mm	47 Nm
M12	10.2 mm	81 Nm

Fabric screws:

Type	Slot dimension	Max torque	Min torque
RTS-TX 5,5x16 Geomet	4.4 mm	5 Nm	6 Nm
RTS-TX 5,5x19 Geomet	4.4 mm	5 Nm	6 Nm
RTS-TX 5,5x22 Geomet	4.4 mm	5 Nm	6 Nm
KFTS-TX 5,5x16 Geomet	4.4 mm	5 Nm	6 Nm

3) Electrical installations are to be made professionally and in conformance with national requirements.
4) Ensure that the door opening conforms to the installation documents that accompany each order and also that the dimensional tolerances stated there, stability and other directions are followed accordingly.

Tools

Recommended installation tools.

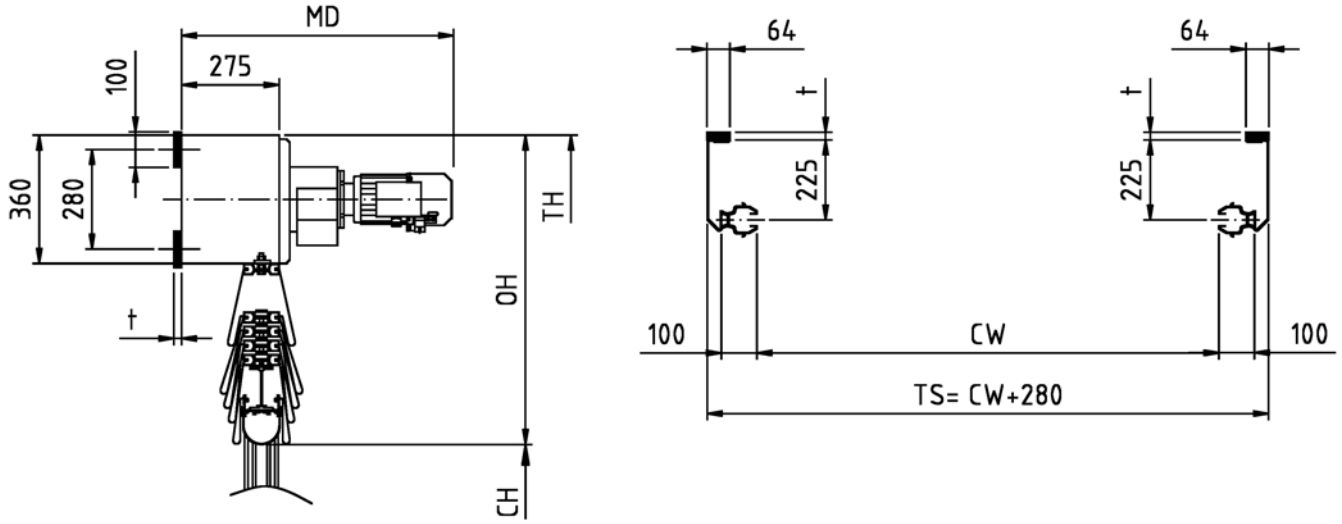
Suitable lifting equipment

To lift personnel: articulated lift, skylift or scaffolding.
To lift material: forklift truck, wheel loader or small crane.

2.2 Installation options

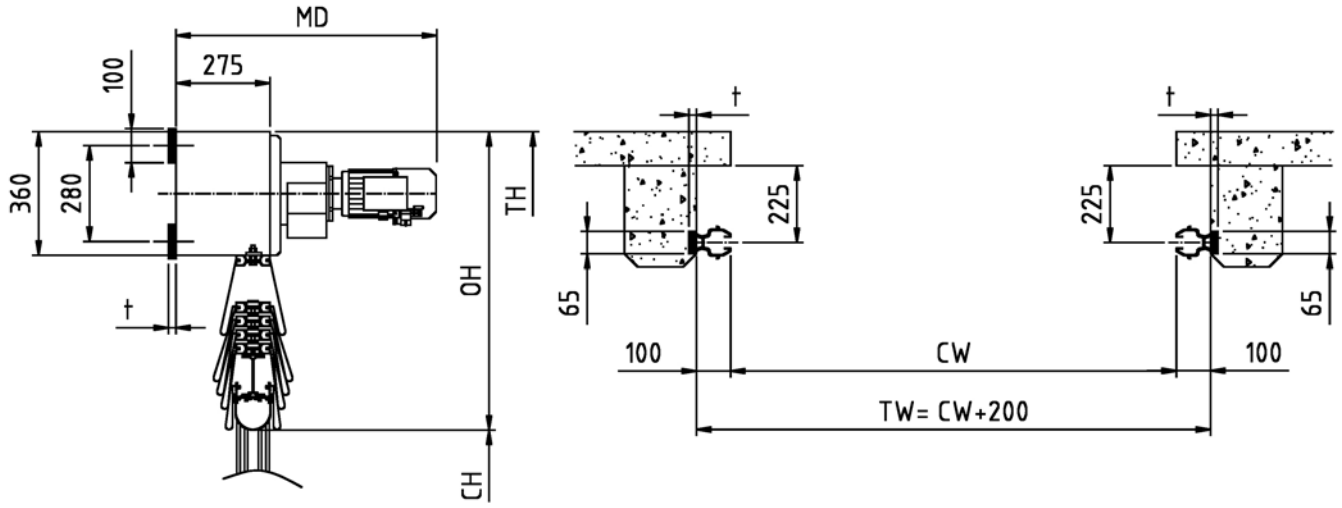
The ASSA ABLOY VL3110 Megadoor vertical lifting fabric door can be installed in accordance with the following options. See installation documents for the delivery.

Header box | Guide rails



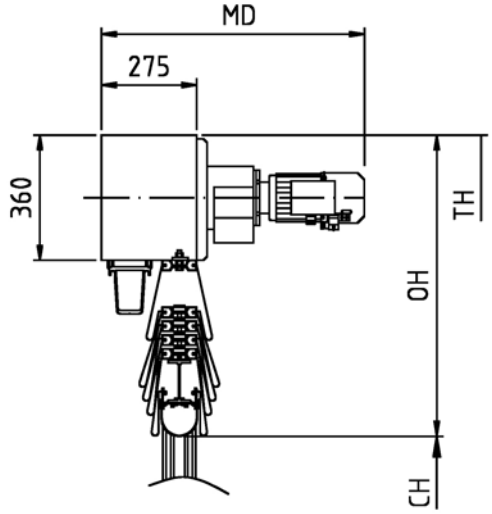
Installation against wall with jambs

Header box | Guide rails

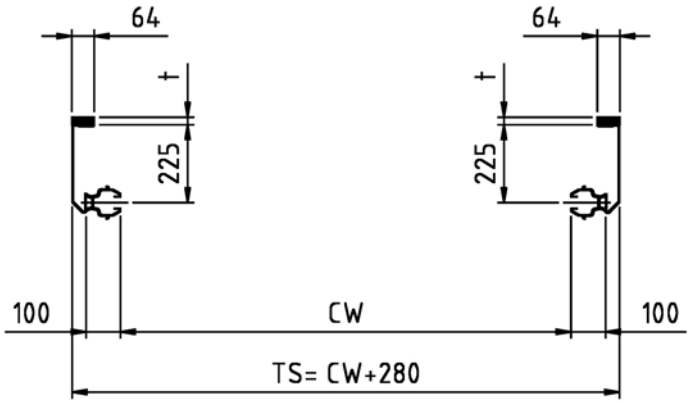


Installation against wall between columns

Header box

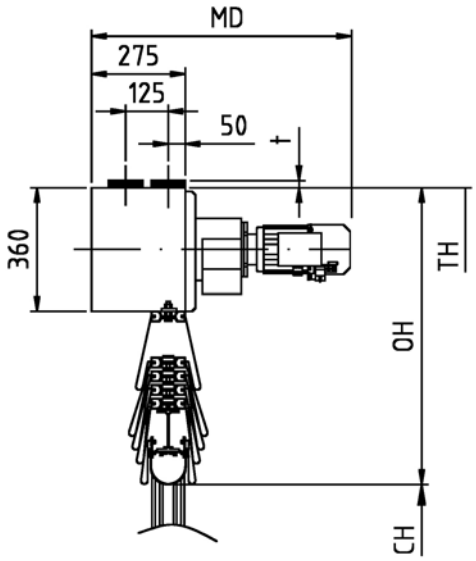


Guide rails

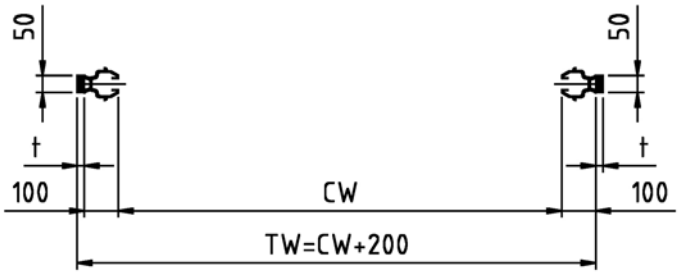


Installation with self-supporting header box and jambs

Header box



Guide rails



Installation in door opening

Explanation:

TH:	Total height	TW:	Total width (± 10 mm)
OH:	Over height	CW:	Clear width
CH:	Clear height	MD:	Motor depth
TS:	Total space requirement (± 10 mm)	T:	Thickness acc. to ”Fastening”

2.3 Guide rails - mounting surfaces

Suitable mounting surfaces must be available to facilitate the fitting of guide rails (see dark fields in the illustration). Mounting surfaces must be:

- Firm and smooth.
- Parallel and deviating not more than 5 mm from the vertical and not more than 2 mm/m in the inward or outward direction from the vertical.

Distance between fixing points should not exceed 1m.

Check for any slope of the floor to determine at which level the fitting of the guide rails is to begin.

The difference in height may not exceed 1 mm/m of the door width.

Fastening:

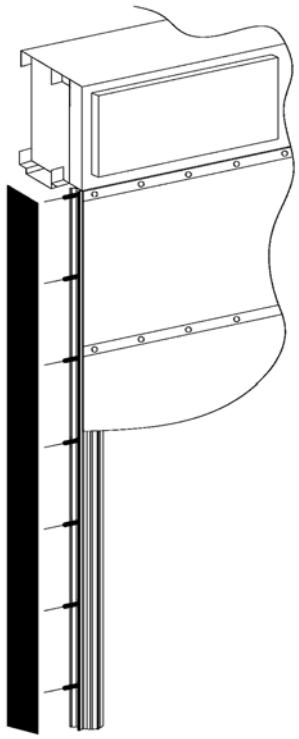
Screw size: M8

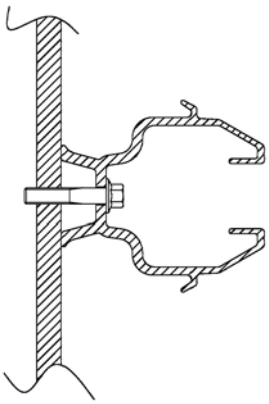
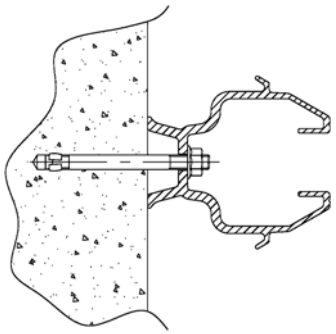
Strength class: 8.8

Fixing surface, steel: t ≥ 6

Fixing surface, concrete: t ≥ 90

See also "Installation options"



Steel:	**Concrete:**
	
Drill holes to thread with M8 tap or self-threading screw M8.	Drill holes for M8 expansion screw.

Do not exceed the recommended tightening torque (applies to both steel and concrete).

2.4 Guide rails - mounting



1) Mark a vertical line for the fixing points.



2) Drill holes for the screws. If the floor slopes, begin at the side that is highest. Then drill screw holes on the other side at the same levels.





3) Begin by fitting the shortest guide rail. Leave about 2 mm between the rail and the floor.

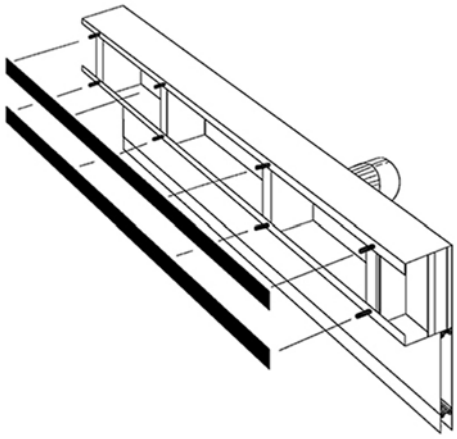


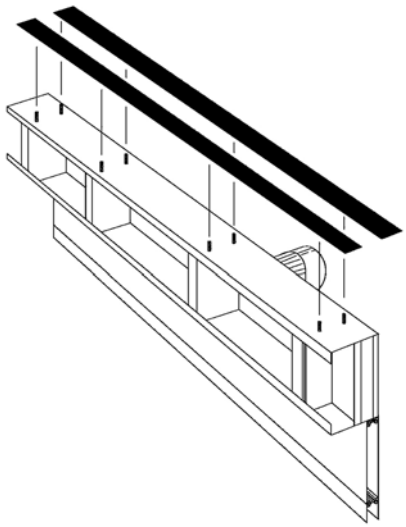
4) Ensure that the guide rails are vertical.



2.5 Header box - mounting surfaces

The rear or top of the header box may be secured to the wall or beams of the building.

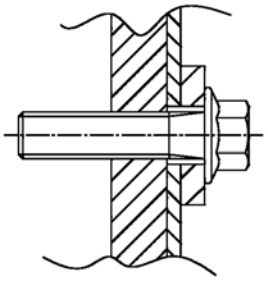




Mounting on wall
Flat, vertical surfaces must be available to facilitate fitting the back of the header box (see dark fields in the figure).

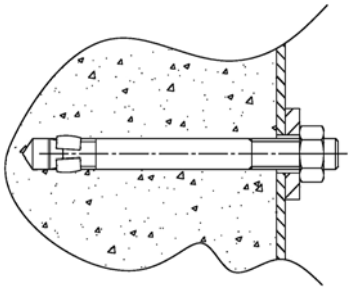
Mounting in opening
Flat, horizontal surfaces must be available to facilitate fitting the top of the header box (see dark fields in the figure).

Steel:



Drill holes for self-threading screw M10 to secure the header box.

Concrete:



Drill holes for expansion screw M10 to secure the header box.

Fastening:
Screw size: M10
Strength class: 8.8
Fixing surface, steel: t ≥ 8
Fixing surface, concrete: t ≥ 100
See also "Installation options"

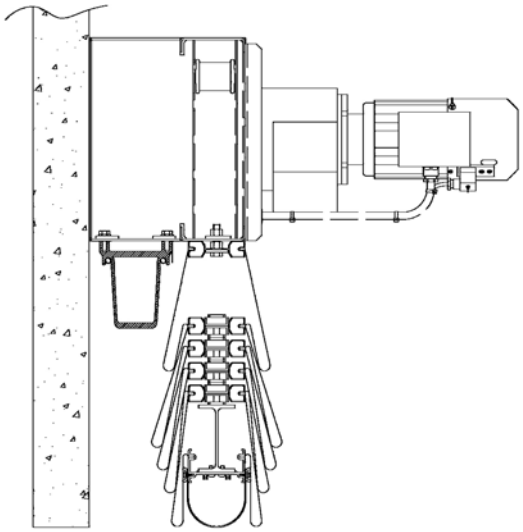
Hole pattern header box fastening:
TW = 140 + 835 + X*1000 + 2*(remaining measure/2) + 835 + 140
Seen from motor side, starting from the left.
X = actual number of 1000 mm distances = int ((TB-2950)/1000)
1000 < remaining measure ≤ 2000

2.6 Header box - self supporting

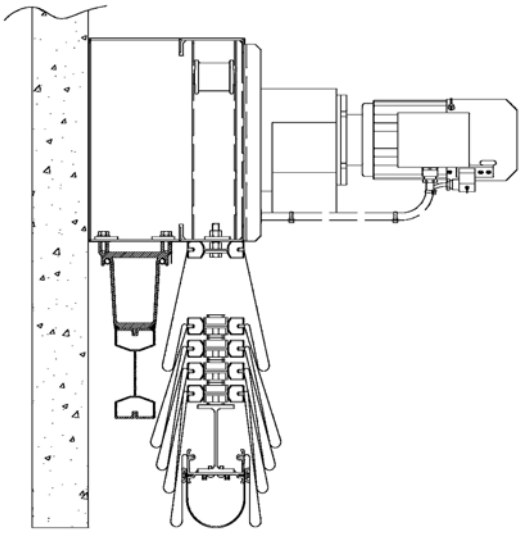
Self-supporting version:

If there is no suitable mounting surface, the door can be delivered in a self-supporting version. The supporting beam, located underneath the header box, is secured by brackets in the building or in the jambs.

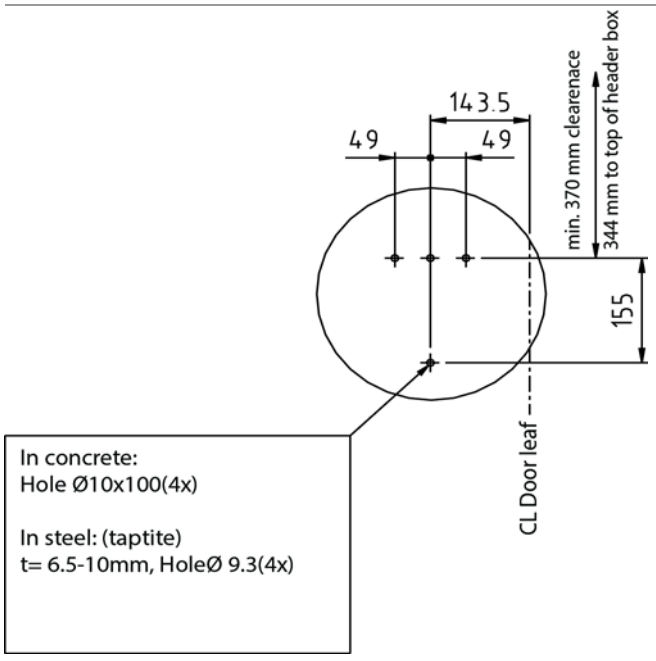
Two variants are available depending on clear width.



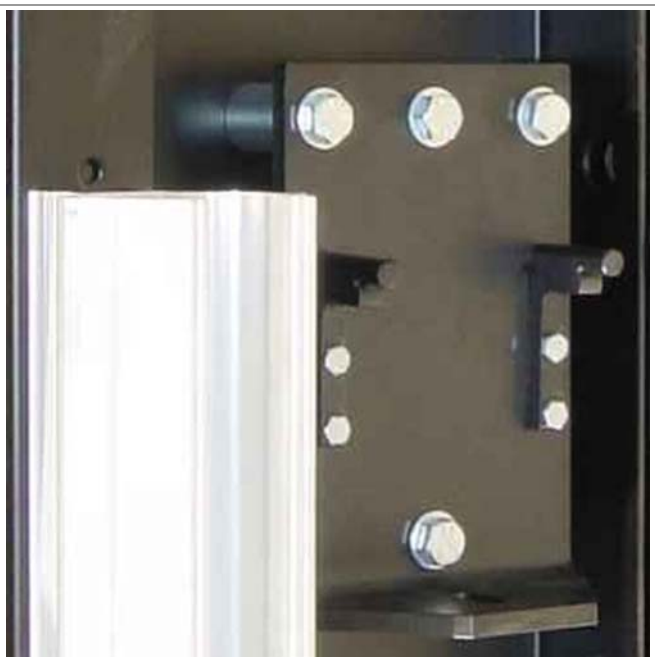
Clear width < 6800 mm



Clear width > 6800 mm



Mounting surfaces in building for supporting beam brackets.



Brackets in jamb for self-supporting version.

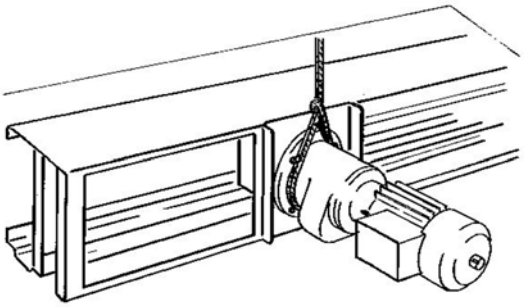
2.7 Header box - mounting



1) Dismantle the header box cladding.



2) Place the gear motor in the correct position on the header box



3) Lift up the header box, either by using straps round the gearing or by lifting underneath the header box (the entire length of the header box must then be supported).

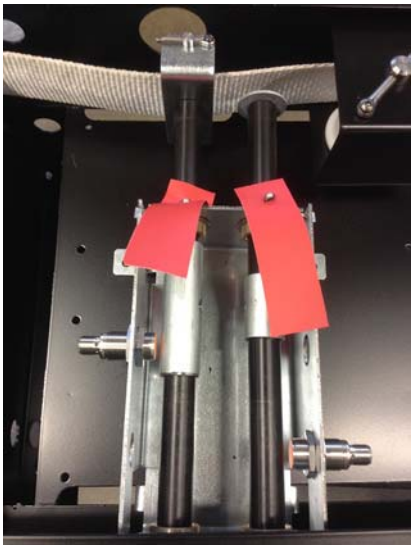


4) Place the header box on top of the already installed guide rails; centrally in relation to the opening. Ensure that the header box is levelled. The difference in height between the ends may not exceed 1 mm per meter width of the header box.

Screw the header box firmly in the holes (M10) or to the brackets in the self-supporting version.



5) Loosen and feed the belt down through the holes in the header box and down in the guide rails.



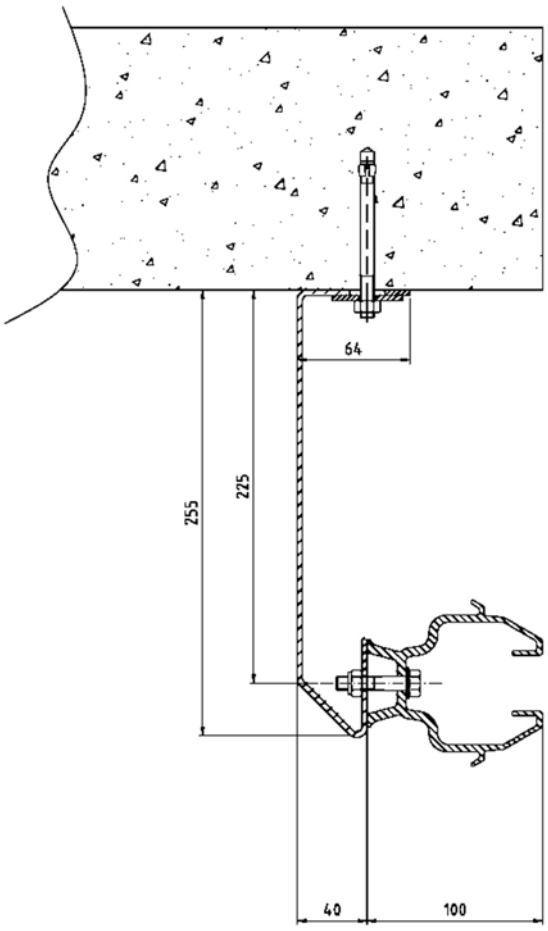
6) Remove pins to release the pushrods in the limit switch boxes.



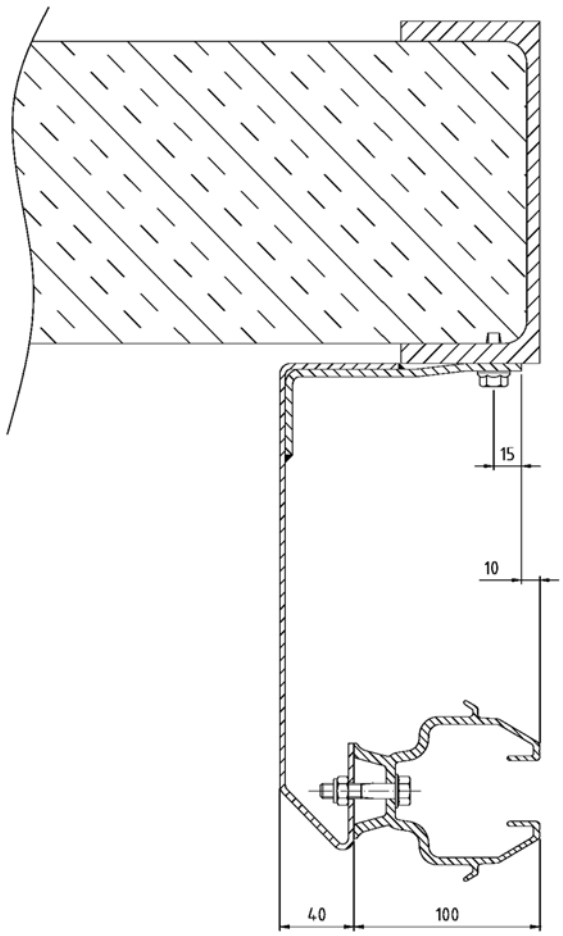
7) Activate the vent plug on the gearbox before use.

2.8 Examples of available options

- Self-supporting header box – see paragraph “Header box, self-supporting”.
- Protective motor casing. In outdoor installation or unclean environment.
- Enclosed “non-motor side”. For installation in opening with the motor facing inwards, the open rear of the header box (facing outwards) must be provided with fixed cladding sheets.
- Jambs – see Installation options. Used when there are no fixing surfaces for the guide rails.
- Extension jamb, making it possible to reach existing steel and still achieve protected positioning of the guide rails.



Jamb



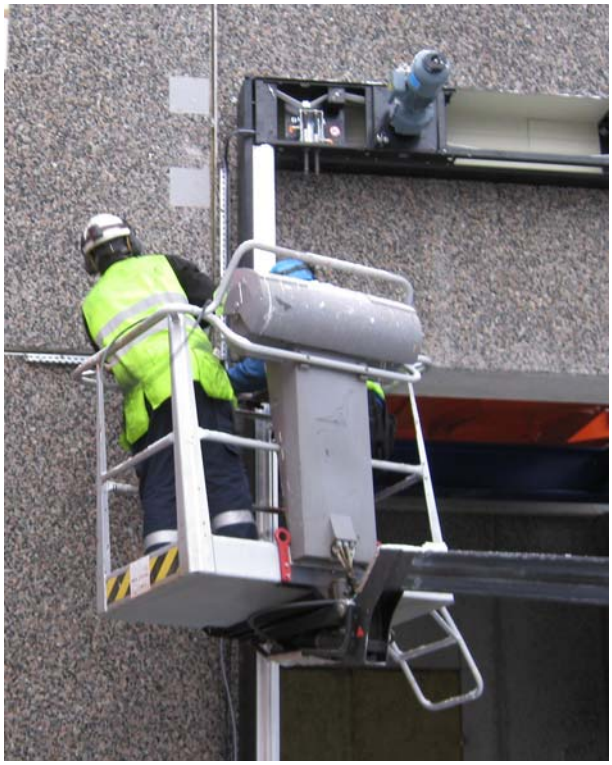
Extension jamb

- Vision panels of different sizes, different types of special fabric.
- Automatic closing function with photocells, magnetic loops, etc. *
- Radio control functions. *
- Etc.

* Instructions and functional descriptions accompany the respective function option

2.9 Control unit and cable





1) Mount the control unit in position according to the documentation. Permanent supply is required to complete the installation according to EU standard. Ensure that the supply fuses are not higher rated than what is approved for the circuit breaker.

2) Connect and pull all cables out of the header box. Separate the data and the power cables by at least 30 cm if possible. Clamp the cables from the header box and downward. Distance between clips about 25 cm; from each bend about 10 cm. Lay cable only vertically and horizontally.

3) Connect cables in control unit according to the wiring diagram.

Do not hammer the cables.

At start-up, all terminal block connections must be retightened to avoid bad contact!

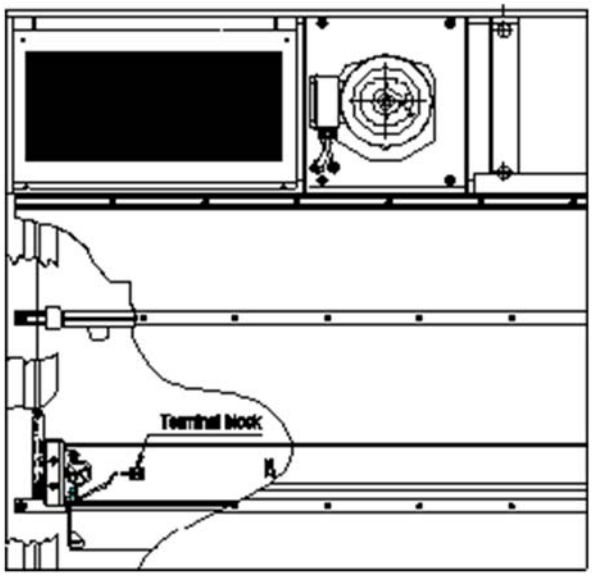
After changes in load center, ensure that the phase order is correct. Run the door with the "Down" pushbutton. If the door goes up, stop it immediately by pushing the Stop or Emergency Stop push-button. Turn off the power supply to the door and then change two phases in the main switch or, if not existing, on the terminal block in the control cabinet of the door.*

*) Not applicable to doors with variable frequency drive

2.10 Bottom section



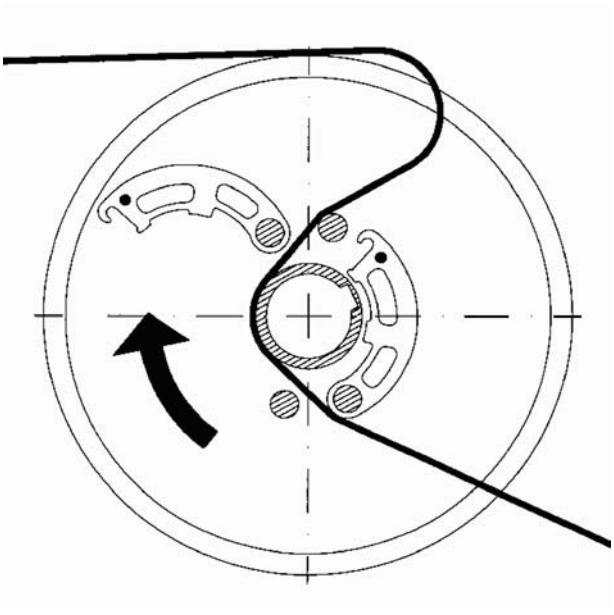
1) Remove the short guide rail



2) Lift in the bottom section, ensuring that the terminal block for the safety-edge cable is to the left of the motor on the motor side.



3) Secure the lifting belts in the safety arresters, ensuring that the retainer ring enters its groove in the shaft. The opening of the retainer ring shall face down.



4) Ensure that the bottom section is horizontal. If not, adjust the belt length by cranking the motor to release the belt segment as illustrated. Use the assembly mode as described in the section "Manual operation".



5) Measure the difference between both ends of the bottom section. Displace the belt half the measured difference towards the higher side of the bottom section.



6) Ensure that the bottom section is now horizontal and sealing against the floor. Fix the belt by turning the belt drum in the direction of the arrow (see illustration above) so that the belt is wound up and locked by the segments.

For test, run the bottom section up into the guide rail. The bottom section shall run smoothly without any noise and tendency to get stuck. Take down the bottom section to the ground after finishing the test and slack the belt.

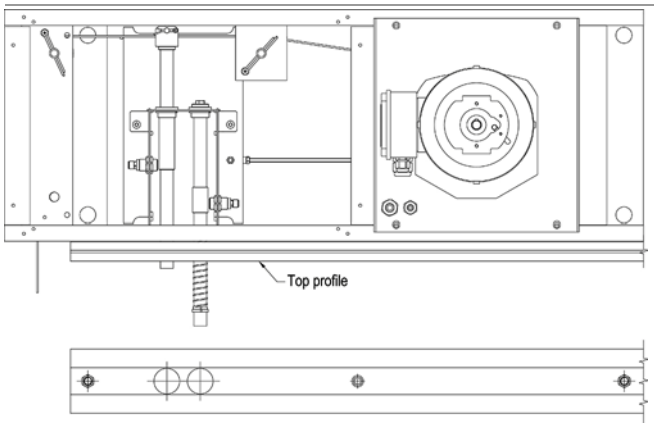
2.11 Door leaf



1) Fit the mounting guides (installation equipment) at the far bottom under the long guide rails.



2) Lift in the door leaf with packing (and pallet if any) and place it on the bottom section. If the door leaf is heavy, attach two straps round it so that it can be lifted into the door opening using a fork lift or by other means.



3) Ensure that the door leaf is facing the right way so that the hole in the top section and fabric is on the correct side (left of the motor on the motor side).



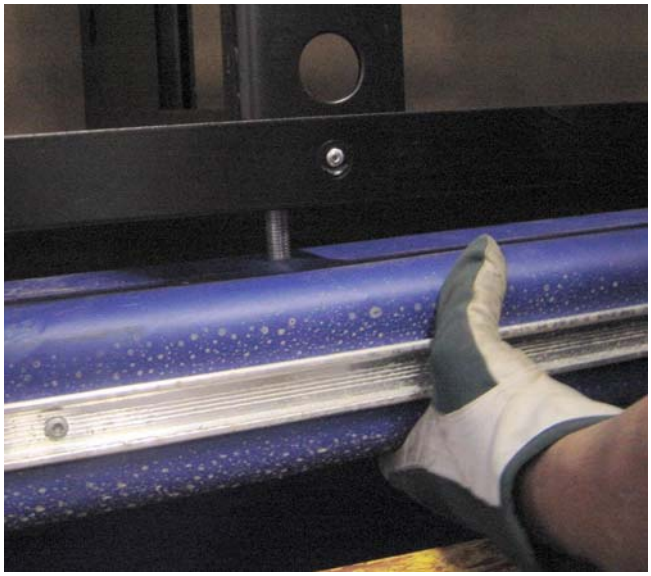
4) Lift up the door leaf and guide it carefully into the guide rails. Lift it enough to make room for the bottom section beneath it. Remove the screws, holding the transport securing devices.



5) Make certain that the fabric, clamp strips and nylon knobs are not damaged.



6) Run the bottom section up to the door leaf. Before it reaches the door leaf, place out blocks on the bottom section where suitable, to create space so that the straps can be removed. When the door leaf is resting on the bottom section, remove the straps and any remaining packing and remove the transport securing devices.



7) Carefully run the door leaf up and stop 200 mm from the bottom of the header box. Crank the motor to lift the door leaf and centre the top section in relation to the header box so that the screws and the push rods get into position.

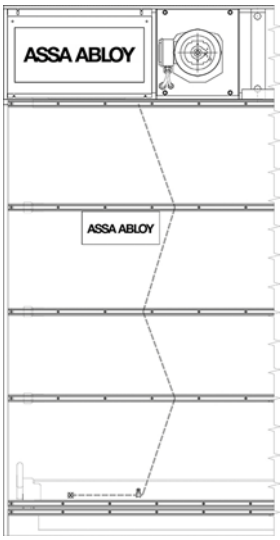


8) Screw the door leaf firmly to the header box and secure the short guide rails.

Do not exceed the prescribed tightening torque. Remove the mounting guides.

2.12 Safety edge cable

Before connecting the safety edge cable, ensure that the power supply to the control unit is disconnected, so that the door cannot be operated unintentionally.

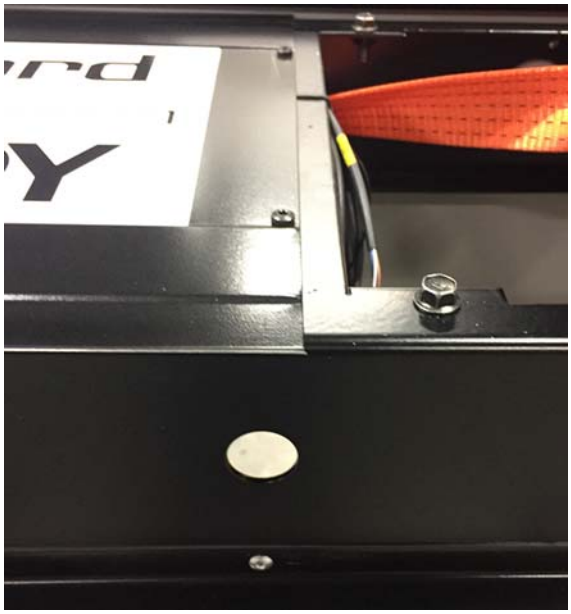




1) The safety edge cable is pre-installed in door leaf at factory.

2) Loosen the cable end that is fixed at the top section.





3) Connect the safety edge cable to the terminal blocks and secure it with the clamp on the bottom section.

4) Thread cable through hole in header box and clamp it downwards from the header box. Connect cable in control unit according to the wiring diagram.

2.13 Safety edge - adjustment

No adjustment is needed for a conductive safety edge.



Conductive safety edge relay.

2.14 Header box - enclosing

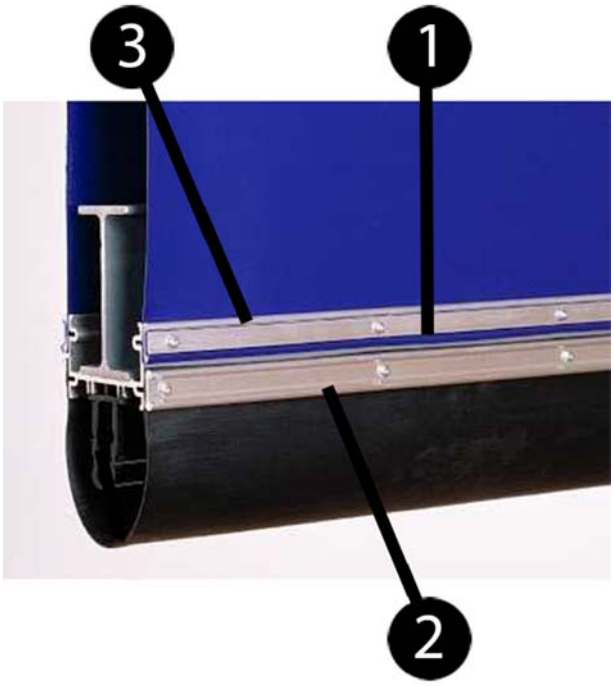




1) Install front cladding and inspection hatches. The cladding must be installed with overlapping. Push the cladding sheets up against the edge of the upper sheet. Seal joints with sealing paste.

2) Fit motor casing if any.

2.15 Fitting fabric to bottom section



1) Clamp strip alt 1
2) Clamp strip alt 2
3) Inter clamp strip alt 3

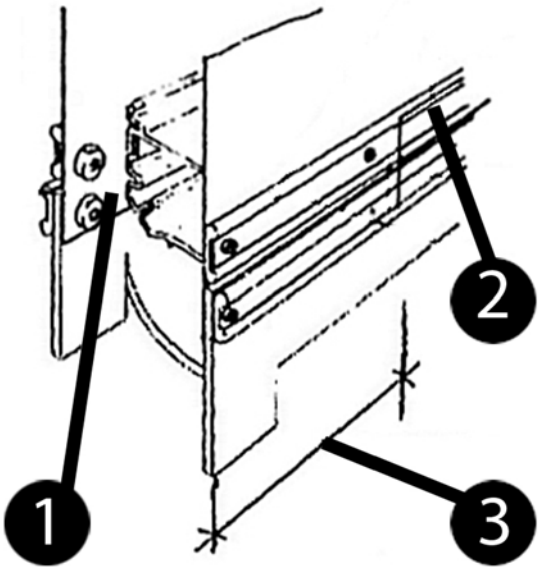
Run the door down so that the floor seal is tight against the floor. Check that the rubber profile and contact strip do not touch the floor, feeling with the hand along the whole width of the door. Distance from floor to screw in clamp strip alt 2 shall be about 110 mm.



1) Remove the upper clamp strips, alt 3 and alt 1. NB. It is important that the clamp strips do not change place when refitting them.



2) Tension the fabric towards the floor and secure the fabric with clamp strip alt 3; beginning at the middle and working outward towards the sides. Ensure that the fabric is stretched and even, especially at the edges.



1) The flap furthest out is to be folded down behind the sealing rubber and secured with screw and nylon knob against the strip and also with the first screw in the bottom section.
2) Between the ends the fabric is to be folded up and secured with clamp strip alt 1 on clamp strip alt 3.
3) Flap 20 mm wider than the distance from edge to the second screw hole.

3) Cut flaps of the fabric at each fabric edge, as illustrated above. Fold up the fabric between the flaps and fasten it with clamp strip alt 1. Loosen the two outermost screws in clamp strip alt 2. Cut the flap 20 mm below clamp strip alt 2. Thread the flap behind the sealing rubber. Secure the fabric in all clamp strips with screw and nylon knob. Refit the other screws in clamp strip alt 2.

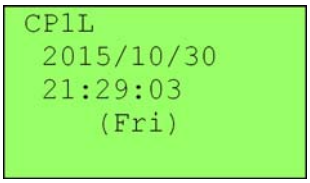
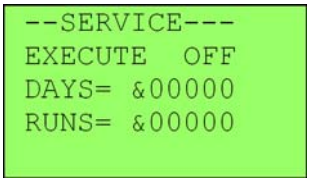
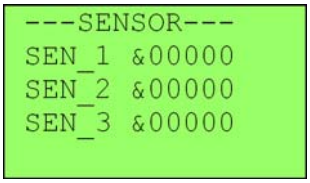
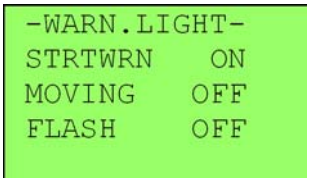
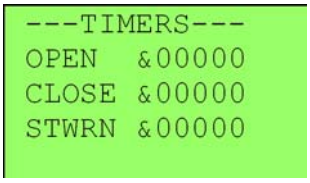
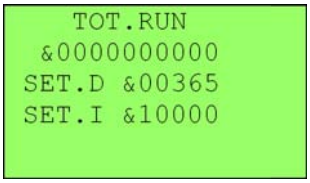


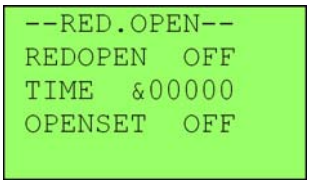
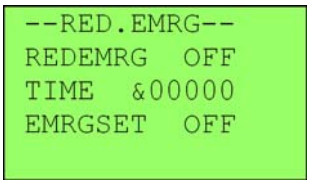
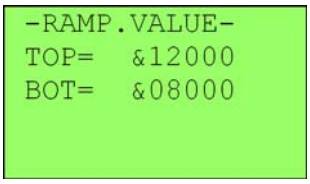
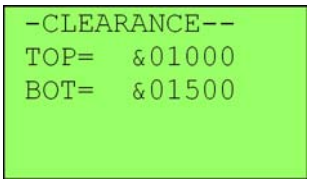
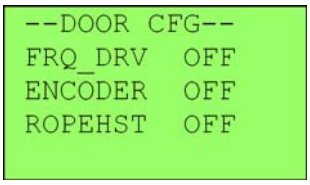
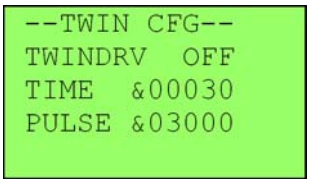
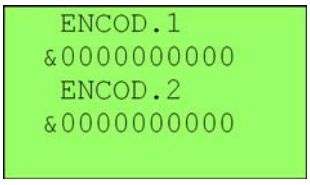
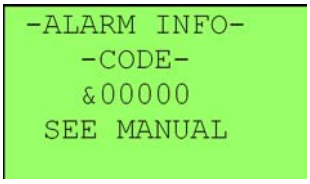
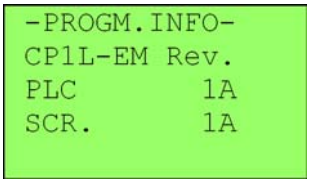
4) Cut away surplus fabric above clamp strip.
5) Configure door operation in the PLC as described in the section "PLC operating instructions".

2.16 PLC operating instructions

2.16.1 Human machine interface

By navigating up or down with the arrow buttons, the below displays can be accessed. To configure the door operation, see the procedure described in "Configuration".

	The PLC start up display with the PLC type, the date, the time and the day.
	Service display. EXECUTE. Service performed. DAYS. Days since last service. RUNS. Runs since last service.
	Safety display. SAFEEDGE. Enable safety edge. SHORTREV. Enable short edge reverse function. OPTOEDG. Enable optical safety edge.
	Sensor configuration display. Set function 00001-00007 of each connected sensor. 00000 Disabled, 00001 Safety, 00002 Automatic opening (timer delayed), 00003 Automatic closing (timer delayed), 00004 Automatic closing (passage triggered, timer delayed), 00005 Alternating automatic opening/closing (timer delayed), 00006 One button function, 00007 Interlock
	Warning light display. STRTWRN. Warning signal before start. MOVING. Warning signal during movement or, if disabled, while in intermediate position. FLASH. Warning signal flashing.
	Timers display. OPEN. Time in seconds to delay open command after activation of the sensor. CLOSE. Time in seconds to delay a close command after deactivation of the sensor. STWRN. Time in seconds to delay start.
	Total runs display. TOT.RUN. Total number of runs. SET.D. Set service interval in days. SET.I. Set service interval in runs.
	Alternative open display. EMRGOPN. Activate emergency open function, door opens when input is OFF. MEMOPEN. Store open command until interlock input is high (cancelled after 3 minutes). SPEC. Special automation feature. Activated with code from electrical schematic.

	Reduced opening display. REDOPEN. Open to a reduced height. TIME. Reduced opening timer in 10ths of seconds from door closed. OPENSET. Set current height to reduced opening. Absolute encoder required.
	Reduced emergency open display. REDEMRG. Emergency open to a reduced height. TIME. Reduced emergency opening timer in 10ths of seconds from door closed. EMRGSET. Set current height to reduced emergency opening. Absolute encoder required.
	Absolute encoder settings display. TOP. Pulse position, relative to fully open, to start ramping down when opening the door. BOT. Pulse position, relative to fully closed, to start ramping down when closing the door.
	Absolute encoder settings display. TOP. Pulse position, relative to fully open, to stop. BOT. Pulse position, relative to fully closed, to stop.
	Door configuration display. FRQ_DRV. Enable variable frequency drive. ENCODER. Enable absolute encoder. ROPEHST. Enable rope hoist.
	Two-motor door configuration display. TWINDRV. Enable two-motor drive. TIME. Skew tolerance timer in 10ths of seconds. PULSE. Skew tolerance timer in pulses.
	Encoder display. Current pulse value of encoder 1. Current pulse value of encoder 2.
	Alarm code display. See paragraph "Alarm codes" in this manual for more information.
	Program information display. CP1L-EM. PLC type. PLC. PLC software version. SCR. LCD software version.

2.16.2 Alarm codes

Navigate through the displays using the UP and DOWN arrow buttons to display "ALARM INFO". Any active alarms will be displayed here. See table below for a description of each alarm code. If there is more than one error at the same time, the display will cycle through them. Inactive alarms can be cleared by restarting the control unit, or by pushing the EMERGENCY STOP in and then releasing it.

Code	Explanation:
&00000	Door ok
&00001	Safety circuit broken
&00002	3-phase limit switch or safety switch actuated
&00003	Absolute encoder communication error
&00004	Absolute encoder direction alarm
&00005	Drive unit alarm
&00006	Control unit temperature alarm
&00007	Contactor fault
&00008	Service alarm
&00009	Self configuration not performed
&00010	Current overload
&00011	Left belt/rope rupture
&00012	Right belt/ rope rupture
&00013	Not in use
&00014	Not in use
&00015	Break resistor temperature alarm
&00016	Illegal position, left side
&00017	Illegal position, right side
&00018	Skew alarm, applicable only for two-motor doors
&00019	Variable frequency brake alarm
&00020	PLC low battery

2.16.3 Manual operation

If the door needs to be operated manually disregarding all sensors and safety features, hold STOP and DOWN for 30 seconds.. Use extra caution while operating in assembly mode. Both OPEN and CLOSE is “hold to run” control. Activating assembly mode resets the stored limits of a door with absolute encoder.

2.16.4 Self configuration

On a door with absolute encoder, to find and store the pulse values for fully open and fully closed respectively, hold STOP and UP for 30 seconds. The door will first close and then open automatically.

2.16.5 Reset service counter

To reset the service counter, push the EMERGENCY STOP, then hold STOP and UP for 30 seconds, then release the EMERGENCY STOP. Alternatively follow the procedure in the "Configuration" section to toggle EXECUTE in the service screen.

2.16.6 Load factory defaults

To load factory default settings in the PLC , push the EMERGENCY STOP, then hold STOP and UP for 60 seconds, then release the EMERGENCY STOP.

2.16.7 Push button shortcuts summary

Assembly mode: STOP + DOWN 30s

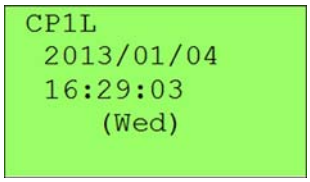
Self configuration: STOP + UP 30s

Reset service counter: EMERGENCY STOP + STOP 30s

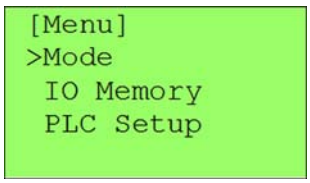
Load factory defaults: EMERGENCY STOP + STOP + DOWN 60s

2.16.8 Configuration

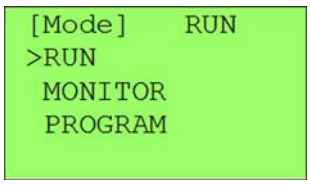
In order to change any parameter the PLC must be in MONITOR mode. The default mode is RUN and it will return to the RUN mode if the control unit is turned off and on. Follow the below example procedure to change to MONITOR mode and enable the short edge reverse function. All other configurable parameters can be changed using the same procedure.



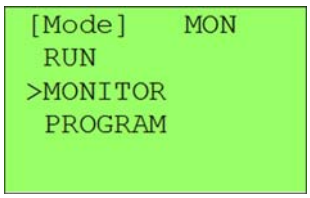
Turn on the power of the control unit and the clock will be shown on the PLC unit.



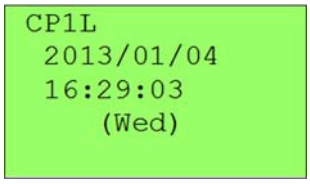
Press the OK + ESC button simultaneously to switch to the main menu.
The line cursor ">" is always displayed on the first line of menu items



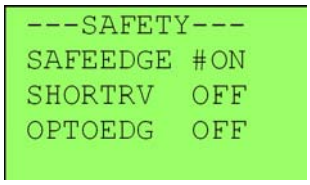
Press the OK button to enter the Mode display.
There is a choice of 3 PLC modes: RUN/MON/PRG.
The line cursor will point to the present PLC mode.
The present mode is RUN.



Press the Down button to select MONITOR.
Press the OK button. The LCD will update the present mode to MON.
Press the ESC button twice to switch to the main menu.
The PLC is ready to have its settings changed.



Press the Down or Up button to move between the displays.



Press the OK + Forward button simultaneously to enter the Change display. The column cursor is at the "#" position.
Press the Down or Up button to select the menu item.



Use the Forward button to move the column cursor to the switch to be set. Use the Down or Up button to select On/Off.
Press the OK button to activate/save the setting.
Press the ESC button to return to the previous display.



When the value is changed, the service counter will be reset. The value returns to OFF automatically.
Press the ESC button to return to the previous display.

2.16.9 Troubleshooting

Problem	Cause	Solution
Door cannot be opened nor closed	Main fuse blown	Replace fuse in distribution panel
	Control circuit fuse blown	Reset fuse in control box
	Emergency button pressed	Release emergency stop button
	Motor overload relay tripped	Reset motor overload relay in the electrical cabinet after it has had time to cool. If overload relay continues to trip, contact the nearest service company
	Limit switch of the crank interlock breaks the control circuit	Ensure that the crank interlock becomes locked in the closed position
	Door is interlocked	Verify sensor settings and connections. The contact of a sensor that is configured for automatic closing must be connected normally closed
	Phase sequence fault (applies only to doors with emergenc power inlet)	Turn off the power supply to the door and change two phases in the main switch
	No power to motor	Verify motor connection, 3-phase limit switch and safety switch where applicable
	Motor break not releasing	Verify incoming and outgoing voltage from the brake rectifier
	Self configuration not performed (applies only to doors with absolute encoder)	Perform self configuration
	Belt rupture limit switch(es) not actuated	Verify integrity of belt and replace if necessary. Verify signal from the belt rupture limit switch(es) to PLC. Use hand crank or enter assembly mode to open door until PLC input indicates signal from belt rupture limit switch(es)
Door cannot be opened	Top limit switch not actuated	Verify function of push rod in limit switch box and signal from the top limit switch to PLC input
Door cannot be closed	Obstacle in guide rail	First, stretch belts by opening door slightly. Then remove obstacle or replace guide rail if it is deformed
	Bottom limit switch not actuated	Verify signal from the bottom liomi switch to PLC input
Door cannot be closed with remote control	Safety edge not working	Verify safety edge and connection cable
Door stops before bottom position after receiving a "Close" impulse command	Safety edge not working. Break in connection cable.	The door can be closed by pressing the push-button "Close", until the door is fully closed. Contact nearest service company for assistance
Door does not close but reverses after receiving a "Close" impulse command	Break in safety edge connection cable	Replace connection cable

2.17 Final inspection and hand over

Final inspection:
See checklist.

Marking:
The door is marked according to EN 13241-1:2003 + A1:2011.

Handing over:
Write an installation report.
Hand over and go through the following documents with the customer:
- User Manual
- Logbook

2.18 Checklist

Door No: _____
Customer: _____
Fitter: _____
Date: _____

General:

Check function of the door, emergency stop and safety edge (reversing function). Ensure that the cables are neatly clamped and not damaged. The door fabric must be clean and tensioned when the door is closed. The door leaf shall run smoothly in the guide rails without any noise.

Header box		Incomplete due to:	Sign.	
OK (x)				
1.	Left belt rupture switch working	_____	_____	_____
2.	Right belt rupture switch working	_____	_____	_____
3.	Top limit switch working	_____	_____	_____
4.	Bottom limit switch working	_____	_____	_____
5.	Instruction plate, manual operation of motor	_____	_____	_____
6.	Crank available	_____	_____	_____
7.	Key in motor shaft	_____	_____	_____
8.	Lifting belt not twisted	_____	_____	_____
9.	Lifting belt does not rub	_____	_____	_____
10.	Lifting belt undamaged	_____	_____	_____
Door leaf				
11.	Fabric undamaged	_____	_____	_____
12.	No screws missing	_____	_____	_____
13.	Rubber seal on bottom profile undamaged	_____	_____	_____
14.	Rubber seal on bottom profile adjusted (cut) against floor at the ends	_____	_____	_____
15.	Safety edge working	_____	_____	_____
16.	Safety arresters working	_____	_____	_____
17.	Safety arresters run freely in the guide rails	_____	_____	_____
18.	Smooth guide rail joints	_____	_____	_____
Electric equipment				
19.	Stop button fitted	_____	_____	_____
20.	Cable clamping completed	_____	_____	_____
21.	Customer instructed regarding functions of the door and user manual handed over	_____	_____	_____
22.	Installation report approved and signed.	_____	_____	_____

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